Problem Set 5

Costs and the Supply Curve

EC306: Intermediate Microeconomics

Instructions. Work through these in class. Show all working and interpret your numerical answers in words. Worked solutions follow at the end.

Question 1 — From technology to a cost function

A firm produces with $q = L^{1/2}K^{1/2}$, facing input prices w (labour) and r (capital).

(a) Set up the cost-minimisation problem and use the tangency condition MRTS $= w/r$ together with the production constraint to derive the *conditional input demands* $L^*(w, r, q)$ and $K^*(w, r, q)$.

(b) Show that the long-run cost function is $C(q) = 2q\sqrt{wr}$. Hence write down long-run MC and long-run average cost, and state what kind of returns to scale this technology has.

(c) Verify Shephard's lemma for labour: show that $\partial C/\partial w = L^*(w, r, q)$, and explain in one sentence why this derivative recovers the cost-minimising input quantity.

(d) The cost function is homogeneous of degree 1 in (w, r). State what this means and give the economic reason it *must* hold for *any* cost-minimising firm, regardless of technology.

Question 2 — Short-run curves and the supply decision

In the short run a firm has total cost $C(q) = \frac{1}{3}q^3 - 4q^2 + 20q + 36$.

(a) Identify fixed cost and variable cost, and derive MC(q), AVC(q), and ATC(q).

(b) Find the output at which AVC is minimised and the value of min AVC. Confirm that MC cuts AVC at that point.

(c) State the firm's shutdown price and write its short-run supply curve $q(p)$ explicitly, including the price region over which it supplies zero.

(d) The fixed cost is 36. Explain why it appears in the firm's *exit* decision but *not* in its short-run shutdown decision.

Question 3 — A shutdown decision under stress

(a) A competitive firm faces $p = 12$. Its cost function gives MC(q) $= 2q$, AVC(q) $= q$, and a fixed cost of 50. Find profit-maximising output, profit (or loss), and state with reasoning whether the firm produces in the short run.

(b) Now suppose this is the firm's persistent situation in the long run. What is its long-run decision, and what is the lowest price at which it would remain in the industry?

(c) **Real data.** Find one documented case (industry report, news, or filing, cited with an access date) of a firm or plant that kept operating at a loss during a downturn, *or* one that shut down. Explain the decision in this course's language: was price above or below the relevant average-cost threshold, and was the cost being escaped fixed or variable?

Question 4 — The long-run envelope and scale

(a) Explain why the long-run average cost curve is the *lower envelope* of the short-run ATC curves, and why LRAC touches each SRATC at only one output (not at the SRATC minimum, except at one special point).

(b) **Real data.** Identify a real industry with a clearly U-shaped or L-shaped LRAC and cite evidence for the shape (e.g. minimum efficient scale estimates).

(c) Connect the shape you found to the likely number of firms the market can support, in 2–3 sentences.

Solutions

Question 1

(a) Minimise $wL + rK$ s.t. $L^{1/2}K^{1/2} = q$. Tangency MRTS $= \dfrac{\mathrm{MP}_L}{\mathrm{MP}_K} = \dfrac{K}{L} = \dfrac{w}{r} \Rightarrow K = \dfrac{w}{r}L$. Substituting into the constraint, $L^{1/2}\left(\frac{w}{r}L\right)^{1/2} = q \Rightarrow L\sqrt{w/r} = q$, so

$$L^\star = q\sqrt{r/w}, \qquad K^\star = q\sqrt{w/r}.$$

(b) $C(q) = wL^\star + rK^\star = q\sqrt{wr} + q\sqrt{wr} = 2q\sqrt{wr}$. Hence MC $= 2\sqrt{wr}$ and long-run average cost $= 2\sqrt{wr}$. Cost is linear in q (and MC = AC, both constant), so the technology exhibits **constant returns to scale**.

(c) $\dfrac{\partial C}{\partial w} = 2q \cdot \frac{1}{2}w^{-1/2}r^{1/2} = q\sqrt{r/w} = L^\star$. This is Shephard's lemma. The derivative recovers the cost-minimising input because, at the optimum, the indirect effect of w through input *re-optimisation* is zero (envelope theorem), leaving only the direct effect = quantity of labour used.

(d) Degree-1 homogeneity in (w, r) means $C(tw, tr, q) = t\,C(w, r, q)$: scaling all input prices by t scales cost by t. It must hold for any cost-minimiser because the optimal input bundle depends only on the price *ratio* w/r; scaling both prices leaves that ratio (and hence the chosen bundle) unchanged, so total cost simply scales by t.

Question 2

(a) FC = 36, VC $= \frac{1}{3}q^3 - 4q^2 + 20q$. MC $= q^2 - 8q + 20$; AVC $= \frac{1}{3}q^2 - 4q + 20$; ATC $= \frac{1}{3}q^2 - 4q + 20 + \dfrac{36}{q}$.

(b) $\dfrac{d\mathrm{AVC}}{dq} = \frac{2}{3}q - 4 = 0 \Rightarrow q = 6$. AVC(6) $= 12 - 24 + 20 = 8$, so $\min \mathrm{AVC} = 8$. Check MC(6) $= 36 - 48 + 20 = 8 =$ AVC(6) — MC cuts AVC at its minimum.

(c) Shutdown price $= \min \mathrm{AVC} = 8$. Setting $p =$ MC: $q^2 - 8q + 20 = p \Rightarrow q = 4 + \sqrt{p-4}$ (the rising branch). Supply:

$$q(p) = \begin{cases} 0, & p < 8 \\ 4 + \sqrt{p-4}, & p \geq 8. \end{cases}$$

(At $p = 8$, $q = 4 + \sqrt{4} = 6$, matching the AVC minimum.)

(d) In the short run the fixed cost 36 is sunk — paid whether or not the firm produces — so it cannot affect the produce/shut decision, which compares p with AVC. In the long run the firm can escape the fixed cost by exiting, so it becomes relevant: the exit test compares p with ATC.

Question 3

(a) $p = 12 = \mathrm{MC} = 2q \Rightarrow q = 6$. With AVC $= q$, VC $= q^2 = 36$. Revenue $= 12(6) = 72$; profit $= 72 - 36 - 50 = \mathbf{-14}$ (a loss). Since $p = 12 >$ AVC(6) $= 6$, the firm covers its variable cost and contributes to fixed cost, so it **produces** in the short run (operating loses 14; shutting down would lose the full 50).

(b) In the long run this persistent loss means $p <$ ATC, so the firm **exits**. It would remain only if $p \geq \min \mathrm{ATC}$. ATC $= q + \dfrac{50}{q}$; $\dfrac{d\mathrm{ATC}}{dq} = 1 - \dfrac{50}{q^2} = 0 \Rightarrow q = \sqrt{50}$, $\min \mathrm{ATC} = 2\sqrt{50} \approx 14.14$. So it stays only if $p \gtrsim 14.14$.

(c) *Model answer / marking notes.* A cited real case (with access date) of a plant kept running at a loss *or* shut down, read in course language: identify whether price sat above AVC (keep operating: cover variable cost, fixed cost sunk) or below it (shut), and whether the cost being escaped was variable (short-run) or fixed/avoidable (long-run exit).

Question 4

(a) The LRAC is the *lower envelope* of the SRATC curves because in the long run the firm chooses the plant size (capital) that minimises cost for each output; every fixed-plant SRATC lies on or above LRAC and touches it only at the single output for which that plant is optimal. That tangency is not at the SRATC minimum except at the one output where LRAC itself is minimised (minimum efficient scale).

(b) *Model answer.* A cited industry with a clearly U-shaped or L-shaped LRAC and evidence on its shape (e.g. a minimum-efficient-scale estimate from an industry/regulatory study).

(c) A U-shape with a small minimum efficient scale relative to market size supports *many* firms; an L-shape or large minimum efficient scale (average cost falling over most of the relevant range) supports *few* large firms and tends toward concentration.